

UNIVERSITY OF COMPUTER STUDIES (TAUNGOO)



BUSINESS MANAGEMENT SYSTEM ADVANCED DATABASE SYSTEM (CS - 4225)

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Group Members

No	Roll No.	Name
1.	UCSTGO - 3083	Mg Naing Ye Htut
2.	UCSTGO - 3128	Mg Kyaw Linn Phyoo
3.	UCSTGO - 3316	Mg Wint Maing
4.	UCSTGO - 3326	Mg Ajay Kumar
5.	UCSTGO - 3083	Mg Min Thatti
6.	UCSTGO - 2926	Mg Min Thant Naung

Supervised By

Daw Khaing Min Kyu

Associate Professor

Faculty of Information Science

University of Computer Studies (Taungoo)

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Abstract

This project presents the design and development of a Business Management System (BMS) tailored for manufacturing operations, with emphasis on Object-Oriented Design & Development (OODD) and Database Systems. The system integrates key modules such as budget and forecasts management, invoices, income and expenses, employees, sales, products, and customer management into a unified platform. Using OODD principles, the system was modeled through use case diagrams, class diagrams, and flowcharts, ensuring modularity and maintainability. The database was designed with entity–relationship diagrams, relation schemas, and a CRUD matrix to guarantee data integrity, security, and efficient operations.

The BMS system supports role-based access, secure data handling, and provides real-time insights through analytical reports. Functional and nonfunctional requirements were addressed to ensure performance, scalability, and reliability. Testing was conducted to validate core features such as login, invoice creation, sales management, and reporting. The successful integration of OODD models with database implementation demonstrates the practical application of theory into a working BMS prototype. This work provides a foundation for future extensions, including mobile access, cloud deployment, and AI-driven forecasting.